GENTSACADEMY · SELF-STUDY FIELD GUIDE

Data Analysis with Spreadsheets and Python

TEC-004 · Week 10 of 10 · IT and Technology

Study independently · keep a notebook · save your evidence

THIS WEEK'S OUTCOME

Use formulas or Python for analysis.

STUDY RHYTHM

6 hours

Short sessions work best: Read · Watch · Make · Reflect.

START HERE · IN PLAIN LANGUAGE

In plain language, this lesson helps you work on use formulas or Python for analysis.. Treat portfolio and review as a skill you can build in small steps: study one idea, compare it with a real Liberian or regional example, and save a short piece of evidence of what you understood.

01
READ
Find the main idea.

02
WATCH
Notice a method.

03
MAKE
Try it locally.

04
REFLECT
Check what changed.

Your learning path

READ FOR THE IDEA

Read for the main idea: Use formulas or Python for analysis.

WATCH FOR THE METHOD

Watch for one method or example connected to portfolio and review.

MAKE EVIDENCE

Apply the idea locally and save analysis notebook and dashboard.

ESSENTIAL QUESTION

How could portfolio and review improve a real situation connected to Data Analysis with Spreadsheets and Python?

Write one sentence before you begin. Revisit it after you apply the activity.

Detailed lesson notes

CONCEPT NOTE

This original lecture note places “Use formulas or Python for analysis.” inside the wider work of Data Analysis with Spreadsheets and Python. The week is not only about remembering a definition. It is about understanding the choices behind the work: clear requirements, repeatable workflow, testing, and safe implementation. Begin with a workplace or service process that could be improved with a digital tool, then use the linked academic source to compare your first idea with a more formal explanation.

A useful way to learn this topic is to move from a situation to a decision and then to evidence. First describe what is happening and who needs a better result. Next identify the information, people, tools, or constraints that shape the decision. Finally make a small, responsible attempt and explain what you would improve. This sequence keeps self-study practical and prevents a learner from copying a solution without understanding the reason for it.

01 · PURPOSE

Purpose: connect portfolio and review to the outcome “Use formulas or Python for analysis.”.

02 · METHOD

Method: define the user, build a small version, test it with safe sample data, and document the result.

03 · EVIDENCE

Evidence: save a short record that another learner could understand and review.

INSTRUCTIONS · WORK IN ORDER

1. Read the guide and linked source for the main concept; write down two unfamiliar terms.
2. Describe a local or regional example without including private or identifying information.
3. Complete the weekly exercise using safe sample data, a fictional scenario, or your own non-sensitive work.
4. Review your result against the self-check questions and record one improvement for the next session.

WEEKLY EXERCISE

Exercise: Data Analysis with Spreadsheets and Python learners should define the user, build a small version, test it with safe sample data, and document the result. Use the week’s focus as your test case and save analysis notebook and dashboard.

Learn from trusted sources

These materials extend this original GentsAcademy guide. They remain owned by the named providers. Read or watch using the source's own access and licence terms.

ACADEMIC READING · SOURCE RECORD

Introduction to Computer Science and Programming in Python

Provider	MIT OpenCourseWare
Kind	University open course
Licence / access	MIT OpenCourseWare terms
Official URL	Open official source at ocw.mit.edu

Use it this week: Read selectively for this week's portfolio and review work.

UNIVERSITY OR PROFESSOR VIDEO · SOURCE RECORD

HTML, CSS, JavaScript

Provider	Harvard CS50
Kind	University lecture or official educational video
Licence / access	Access terms apply; video remains the property of its provider and platform.
Official URL	Open official source at youtube.com

Use it this week: Watch with the Data Analysis with Spreadsheets and Python learning outcome in mind; take notes on an idea to test locally.

Turn study into skill

PRACTICE LABORATORY

Submit and present the Data Analysis with Spreadsheets and Python capstone evidence.

Keep your work simple, local, and real. A basic note, recording, checklist, screenshot, calculation, or plan is valid practice.

EVIDENCE CHECKLIST

- □ I named the main idea.
- □ I noted one example.
- □ I tried one small action.
- □ I saved analysis notebook and dashboard.

SELF-CHECK · BEFORE YOU FINISH

1. What is the main idea in your own words?
2. Which decision or assumption most affects the result?
3. What evidence would show that your approach is useful and responsible?
4. What is one next action I can test or improve?

REFLECT AND CONNECT

In 3–5 sentences, explain how this week's concept could improve a real organisation, community service, workplace, or learner project. Identify one assumption, one risk, and one action you would test next.

Vocabulary + notebook

Use this final page as a learning checkpoint. Clear terms and brief notes make independent study easier to continue.

SMALL SELF-STUDY GLOSSARY

self-directed study (self-study) means you manage your own learning plan. learning evidence (evidence) is the proof you save. A learning source (source) is credited to its original provider.

Glossary reference list

***evidence* – learning evidence**: A saved note, plan, calculation, recording, checklist, or other item that shows what you can do.
***self-study* – self-directed study**: A learner plans, studies, practises, and checks understanding without waiting for a live class.
***source* – learning source**: An external reading or video credited to its original provider.

MY NOTE BANK

The one idea I want to remember:

The example I will look for this week:

FINISH WELL

Choose a time for your next study session. Small regular sessions are more useful than waiting for a perfect long study day.

SHARE SAFELY

Discuss an idea with a trusted peer, mentor, or colleague. Keep your own notes and do not share sensitive personal or workplace data.